Samitha Bandara

Final-Year Engineering Undergraduate | AI Researcher

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Research Profile

Final-year Electronic and Telecommunication Engineering undergraduate researching efficient multimodal AI, visual representation reuse, 3D/4D perception, and learning-based robotic control. Interested in resource-aware models that perceive, reason, and act effectively, with the long-term goal of pursuing a PhD and an academic research career.

Research Experience

Research Assistant

Singapore Management University, School of Computing and Information Systems — Singapore

- Investigated efficient vision-language models for repeated and multi-turn visual interactions through visual information reuse and compression.
- Explored Discrete Cosine Transform (DCT) based spectral compression, layer-wise reconstruction adapters, learned selection mechanisms, and visual attention or KV-cache reuse.
- Conducted experiments with multimodal models and visual question-answering benchmarks, focusing on representation efficiency and end-to-end inference cost.
- Contributed to two manuscripts under review at a leading international AI conference: first author on one and co-author on another.

Research supported through the SMART Mens, Manus, and Machina (M3S) programme. Supervised by Prof. Archan Misra, with close guidance from Dr. Dulanga Weerakoon.

Current Research Projects

Representation Reuse for Point-Cloud Vision-Language Models

Exploring transformation-aware reuse of visual representations across dynamic point-cloud scenes to reduce redundant processing as viewpoints and scene geometry evolve.

Learning-Based Stabilization for an Aerial Manipulator

Final-year project developing reinforcement-learning-based stabilization for a simulated aerial platform with a robotic arm, using PPO and Gazebo. Research collaboration with Prof. Rajat Talak, National University of Singapore.

Education

BSc Engineering Honours in Electronic and Telecommunication Engineering — Expected 31 May 2027

University of Moratuwa, Sri Lanka — CGPA: 3.79 / 4.00

- CGPA calculated across 110 completed GPA credits; Dean's List for Semesters 1, 2, and 3.
- Relevant coursework: Pattern Recognition, Image Processing and Machine Vision, Data Structures and Algorithms, Linear Algebra, Applied Statistics, Signals and Systems, Electronic Control Systems, and Communication Networks.

GCE Advanced Level - Physical Science — 2022

Nalanda College, Colombo, Sri Lanka — 3 A grades

Z-score: 2.3862 | Island rank: 278 | Combined Mathematics, Physics, and Chemistry.

Research Interests

Efficient deep learning | Vision-language and multimodal models | 3D/4D perception | Robot perception and intelligence | Reinforcement learning | Computer vision | World models | Mathematics and signal processing for machine learning

Selected Technical Projects

Transformer Optimization GitHub
Explored LoRA/PEFT adapters, 4-bit and 8-bit quantization, KV caching, and practical efficiency techniques for transformer models.

CoreLlama - Llama 2 from Scratch GitHub
Implemented RMSNorm, Rotary Positional Embeddings, Grouped Query Attention, KV caching, and SwiGLU directly in PyTorch.

VLM-gamma - Multimodal Vision-Language Model GitHub
Built a multimodal model from scratch using a Vision Transformer encoder, transformer decoder, and Grouped Query Attention.

gpt2-lite - GPT-2 Reproduction GitHub
Reproduced GPT-2 with a custom training pipeline covering mixed precision, compiled execution, optimized attention, gradient accumulation, scheduling, validation, and BPE tokenization.

nanoViT - Vision Transformer from Scratch GitHub
Implemented patch embeddings, multi-head self-attention, transformer blocks, and image-classification pipelines.

MLP Character-Level Language Model GitHub
Implemented a character-level neural language model with backpropagation, batch normalization, Kaiming initialization, training, and evaluation.

Academic Engineering Projects

Battery Profiler - STM32 Embedded System GitHub
Designed a PCB with voltage and current sensing and a MOSFET load; implemented STM32 firmware for real-time acquisition and battery characterization.

Analog Five-Band Equalizer GitHub
Designed frequency-selective analog filter stages, gain controls, and a tested PCB prototype for real-time audio equalization.

Technical Skills

Programming	Python, C, C++, MATLAB
Machine Learning	PyTorch, Hugging Face, TensorFlow, NumPy, Pandas, OpenCV
Research Topics	Vision-language models, transformers, computer vision, reinforcement learning, 3D/4D perception, model compression
Systems	Linux, Slurm, Docker, GPU training and optimization, Git
Robotics	Gazebo, PPO, simulation-based control, robotic perception
Embedded Engineering	STM32, embedded C, PCB design, sensors, data acquisition
Documentation	LaTeX, technical writing, Microsoft Office
Languages	English, Sinhala

Leadership and Activities

- Group leader for the final-year aerial-manipulator research project; responsible primarily for the reinforcement-learning component.
- Prefect, Nalanda College (2017–2019); participated in athletics, football, the Interact Club, and the Photographic and Art Society.
- Member of the University of Moratuwa Nature Team.